

AWS Cloud Security & Migration Lab

Technical Portfolio Report | 2026

1. Objective

Objective: Build a small AWS migration-style environment and apply practical security controls across networking, compute, storage, identity, and audit logging. The lab connected cloud infrastructure concepts to real security operations rather than treating each AWS service as an isolated definition.

2. Environment Built

Component	Implementation
Network	VPC 10.10.0.0/16; two public and two private subnets across two Availability Zones.
Connectivity	Internet Gateway for the public subnet; no NAT Gateway deployed.
Private service path	S3 Gateway VPC Endpoint.
Compute	Amazon Linux EC2 instance running Apache HTTP Server.
Security Group	HTTP TCP/80 public; SSH TCP/22 restricted to one administrator /32 source.
Storage	Private S3 bucket with Block Public Access enabled.
Identity	EC2 IAM role using temporary credentials; no static access keys stored on the instance.
Authorization	Bucket-specific read policy using s3:ListBucket and s3:GetObject.
Audit	CloudTrail Event History reviewed for AWS API and management activity.

3. Architecture

```
Internet
|
Internet Gateway
|
VPC 10.10.0.0/16
|-- Public Subnet --> EC2 Web Server --> Apache HTTP
|                               |-- Security Group: 80 public / 22 admin /32
|                               |-- IAM Role --> S3 Endpoint --> Private S3
|-- Private Subnets (no direct Internet Gateway route)
AWS API activity --> CloudTrail Event History
```

4. Security Controls Implemented

- Network segmentation: separated public and private network space; private subnets have no direct Internet Gateway route.
- Restricted administration: SSH limited to one administrator public IP using a /32 rule.
- Private storage: S3 Block Public Access remained enabled; anonymous object access returned AccessDenied.
- Role-based identity: EC2 used an IAM role and automatically received temporary AWS credentials.
- Least privilege: broad S3 read access was replaced with a bucket-specific ListBucket/GetObject policy.

5. Validation Performed

- Connected to EC2 using SSH key authentication.
- Installed and enabled Apache; reached the test page from the Internet over HTTP.
- Confirmed anonymous access to the S3 object was denied.
- Used aws sts get-caller-identity to verify an assumed IAM role.
- Listed and read the private S3 object from EC2 using the role.
- Re-tested S3 access after replacing broad permissions with the least-privilege policy.
- Reviewed CloudTrail Event History and the fields used to investigate AWS control-plane actions.

6. Production Security Review

Area	Production Direction
Transport	Use HTTPS/TLS on TCP 443 instead of plaintext HTTP application delivery.
Administration	Prefer no public SSH; use Systems Manager Session Manager, VPN, bastion, or another controlled path.
Workload placement	Keep backend resources private where practical; consider a load balancer for Internet-facing delivery.
Data protection	Maintain private S3 access; add appropriate encryption, versioning, recovery, logging/monitoring, and lifecycle controls.
Operations	Centralize logs and alerts; establish patch and vulnerability management for EC2 and applications.

7. Migration Concepts Reviewed

The build was mapped to common migration patterns and services: Rehost, Replatform, Refactor, AWS Application Migration Service (MGN), AWS Database Migration Service (DMS), AWS DataSync, Site-to-Site VPN, Direct Connect, and landing-zone design. These migration services and hybrid-connectivity options were reviewed conceptually and were not deployed in this lab.

8. Outcome

The lab demonstrated how AWS networking, EC2, IAM, S3, Security Groups, and CloudTrail work together as one security architecture. The strongest practical outcomes were role-based access without static credentials, least-privilege S3 permissions, validation of private object access, and troubleshooting source-based SSH access through Security Group rules.